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Research paper

# Contextual reinforcement learning for market making via multi-task self-supervised learning

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ABSTRACT

Market makers play important roles in modern financial transactions by simultaneously providing buy and sell limit orders to offer liquidity and earn the bid–ask spread. In recent years, several deep reinforcement learning techniques have been developed for high-frequency quantitative trading tasks such as market making. However, due to the uncertainty and variability of financial markets, reinforcement learning for trading needs to overcome two major challenges: concept drift and insufficient generalization ability. In this study, we propose a novel Contextual Reinforcement Learning framework for training a generalized market-making trading strategy, which utilizes financial market sentiment indicators as contextual information, enabling reinforcement learning agents to adjust their order placement decisions according to different market contexts. Subsequently, we introduce a Self-Supervised Market Context Network that incorporates multiple auxiliary tasks with financial domain knowledge, which can further enhance the agent's sensitivity and understanding of the market environment. Furthermore, we adopt the replay-based method from Continual Learning, performing policy fine-tuning by selecting appropriate historical environments, effectively addressing issues of concept drift and catastrophic forgetting. We conducted experiments on real-world tick-level data from the Taiwan Stock Exchange Capitalization Weighted Stock Index (TAIEX) options. Under the same inventory risk, our strategy consistently outperformed state-of-the-art baselines in terms of several evaluation metrics, demonstrating superior profitability per unit of risk. This demonstrates that our proposed framework enables the model to adapt to out-of-sample markets and can also avoid risk and achieve steady profits even in extreme market environments.

## 1. Introduction

Market making is a high-frequency trading strategy that aims to earn the bid–ask spread while providing liquidity to market participants. When conducting market-making activities, market makers place limit ask and bid orders, waiting for counterparts to trade, and earn profits through the bid–ask spread between the prices. However, when only one side of the order (i.e., ask or bid) is executed, market makers will keep buying (selling) the asset when the price goes down (up), the risk arising from inventory accumulation is called inventory risk. Market makers are supposed to minimize their inventory via adjusting the spread and skew of quotes dynamically, and therefore, it is really challenging to build such a strategy that can sustainably make a profit while controlling the inventory risk.

### 1.1. Background

Traditionally, market making is formalized as a stochastic optimal control problem, which is basically studied in the financial engineering and statistics domains. Ho and Stoll (1981) proposed an optimal strategy of a single dealer through a continuous time Poisson jump process, bringing inspiration to the later research. For example, Avellaneda and Stoikov (2008) designed a quoting strategy (AS model) through the estimation of mid price and spread, inventory, and terminal time are also considered in this work. Later on, more research based on the AS model, including additional features, is proposed. However, the assumptions about market behavior are relatively naive, as they are not practical to apply to the real market, nor can they have good adaptability to the new market environment. To solve the above problems, deep

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reinforcement learning (DRL) is applied to market-making problems in an attempt to find more complex trading strategies from historical or simulated data.

Recent advancements in DRL have made it a powerful tool for financial trading and market-making. DRL agents can adapt strategies through interaction with the market, process high-dimensional data such as prices, volumes, and investor sentiment, and optimize towards complex objectives like return maximization and risk control. However, existing strategies primarily focus on modeling high-frequency order books; they often neglect the influence of long-term market trends and sentiment. We believe that short-term trading behavior is also influenced by the current sentiment of the market. For instance, when the market participants are greedy, there may be more buy orders on that day. Lack of considering long-term market information could compromise the model's ability to adapt its quoting strategies in different market scenarios. To mitigate this, we need to incorporate more market information into our model, such as the put–call ratio or open interest in derivative markets. These indicators reflect investor sentiment and are critical sources of information for understanding the overall market condition. Therefore, when the agent is making decisions, it needs to consider not only short-term order book changes but also comprehend and account for the current overall market conditions. The strategy combining short-term and long-term observations can enhance the model's adaptability and improve its generalization ability in fluctuating market environments.

### 1.2. Motivation

To address the limitations of existing DRL-based market-making models, we are motivated to explore a more generalizable and robust framework. Contextual Reinforcement Learning (CRL), a recent development in RL research, emphasizes incorporating latent environmental information to improve agent adaptability. We believe this concept can be effectively transferred to financial markets, where market sentiment, investor positioning, and macro indicators form a latent context. Integrating such context allows the agent to adjust quoting strategies more effectively under different market regimes. However, raw financial time series data – such as price returns, open interest, and volatility indices – are often noisy, high-dimensional, and non-stationary. This creates challenges for stable learning and increases the risk of overfitting. To extract meaningful representations from these complex signals, we adopt Self-Supervised Learning (SSL) techniques. Specifically, we design auxiliary tasks based on future market sentiment indicators, allowing the model to learn compact, informative market context embeddings aligned with the target trading task. Moreover, market conditions evolve over time. Concept drift poses a serious threat to agent performance, especially in high-frequency settings. Models trained on historical data may degrade when deployed in new environments. To mitigate this, Continual Learning (CL) becomes essential, enabling agents to adapt without forgetting previously learned knowledge. Instead of naively retraining on new data, we propose a selective replay-based approach that helps the agent retain past experience while adapting to changing market regimes. These motivations lead us to propose a framework that combines contextual reasoning, SSL, and continual adaptation, aiming to achieve stable and adaptive market making.

### 1.3. Contributions

In this study, we introduce a novel Contextual Reinforcement Learning framework for Market Making (MM-CRL), which enables the agent to integrate market context into the decision-making process, thereby maintaining competitiveness in an increasingly dynamic market environment. To accurately represent the market context, we employ a Self-Supervised Market Context Network (SSMCN) with multiple auxiliary tasks, which can effectively embed market information into a representative low-dimensional market context. A Continual Learning (CL) mechanism is also designed in our framework to mitigate the problem of concept drift, which helps our agent better adapt to the current market and become more stable in a dynamic environment. Finally, we conduct our experiments on a real-world tick-level dataset of Taiwan Index Options (TXO) from TAIFEX. The experimental results show that our method has better profitability under the same inventory risk.

Our contributions are listed as follows:

- We propose a novel Contextual Reinforcement Learning framework for market making (MM-CRL) to enhance the agent's generalizability.
- A Self-Supervised Market Context Network (SSMCN) with multiple auxiliary tasks is implemented to extract features from heterogeneous market sentiment indicators.
- Continual learning mechanism is applied for out-of-sample environments to improve the agent's adaptability and address the issue of concept drift.
- The experimental results show that our method has a better profitability under the same inventory risk.

## 2. Related work

### 2.1. Evolution of market making methods

In recent years, there has been increasing interest in applying DRL to optimize market-making strategies, as they are capable of capturing complex patterns in market data and often achieve superior risk-adjusted returns compared to analytical methods (Gašperov et al., 2021). Early methods, such as the Avellaneda-Stoikov (AS) model (Avellaneda and Stoikov, 2008), relied on stochastic control and strong assumptions about order arrival processes. Spooner et al. (2018) introduced an asymmetrically dampened PnL-based reward function to discourage speculation, and employed manually engineered state representations using a linear combination of tile codings. Ragel (2024) later addressed a key limitation in previous approaches by proposing a data time travel method to simulate the market impact of agent actions more realistically. Carlsson and Regnell (2022) conducted a comparative analysis between analytical and RL-based market-making strategies using tabular Q-learning and Double DQN, discussing their trade-offs. More recently, Guo et al. (2023a) proposed ORL4MM, the first offline reinforcement learning framework for market making, which leverages historical trading data to improve training stability and efficiency, followed by online fine-tuning to adapt to new market conditions.

Recent DRL-based methods increasingly adopt deep neural networks for market representation. For example, Sun et al. (2022a) applied a Deep Recurrent Q-Network (DRQN) with LSTM units for capturing temporal market patterns, while Guo et al. (2023b) introduced Attn-LOB, a CNN-attention model for extracting features from noisy, high-dimensional limit order book (LOB) data. As a further development of DRL-based AS strategies, Ekarittikrai and Rattagan (2024) emphasized quoting decision optimization, allowing agents to dynamically determine whether to quote or not, rather than following a static quoting policy. In addition, Chen and Huang (2021) demonstrated how multimodal DRL combined with sentiment analysis can influence trading decisions by incorporating multiple data sources to enhance financial market predictions. Multi-agent reinforcement learning approaches have gained attention for their ability to simulate more complex financial interactions. Lin et al. (2022), Chen et al. (2023) applied multi-agent systems to financial markets, demonstrating their effectiveness in managing dynamic environments and improving trading strategies through enhanced learning architectures. He et al. (2023) introduced a multi-agent virtual market model that improves the generalization of RL-based trading strategies, showing the effectiveness of multi-agent approaches in complex financial environments.

**Table 1**
Comparison of SOTA methods.

| Method | Techniques | Context modeling | Continual learning | Representation learning | Application |
|---|---|---|---|---|---|
| Spooner et al. (2018) | Temporal-difference Learning | X | X | X | Market Making |
| AS (Avellaneda and Stoikov, 2008) | Stochastic Optimal Control | X | X | X | Market Making |
| Alpha-AS (Falces Marin et al., 2022) | DDQN, AS model | X | X | X | Market Making |
| CMDP (Hallak et al., 2015) | Contextual MDP, CECE (Cluster-Explore-Classify-Exploit) | V | X | X | Recommenda-tion |
| Attn-LOB (Guo et al., 2023a) | Dueling DQN, PPO | X | X | CNN, Attention | Market Making |
| DRLMM (Sun et al., 2022a) | DRQN | X | X | LSTM | Market Making |
| ORL4MM (Guo et al., 2023b) | Offline RL | X | X | CNN, Attention | Market Making |
| Chen and Huang (2021) | Multimodal DRL, Sentiment Analysis, Influence Model | V | X | MDRNN (Multimodal DRNN) | Single-asset Trading |
| Chen et al. (2023) | Knowledge Distillation, RAMA(Role-Aware Multi-Agent) | V | X | Distillation | Portfolio Management |
| MVMM (He et al., 2023) | DDRL, Multi-Agent Virtual Market Model | V | X | DRNN | Single-asset Trading |
| MM-CRL (Ours) | DDQN, Contextual RL, Self-Supervised Learning | SSMCN | V | TCN, SSL | Market Making |

Table 1 summarizes key differences among representative market-making methods, focusing on used techniques, context modeling, continual learning, and representation learning. Traditional methods like AS and Alpha-AS lack adaptability to changing environments. More recent approaches, such as DRLMM and ORL4MM, introduce deep RL, but omit explicit context modeling or continual adaptation. MVMM simulates diverse scenarios via GANs for better generalization. Recent multi-agent frameworks, such as the RAMA-based teacher–student distillation model, incorporate role-aware designs to improve decision-making in volatile markets. However, these approaches typically require strong domain priors or rely on a synthetic training setup, which may limit their portability or require extensive domain engineering across markets. In contrast, our proposed MM-CRL uniquely combines self-supervised context modeling (SSMCN), continual learning, and representation learning to improve robustness in dynamic market conditions. This integrated design enables our method to adapt to unseen environments and maintain stable performance over time.

### 2.2. Self-supervised learning

In recent years, Self-Supervised Learning (SSL) has become a significant and popular method due to its superior performance in unsupervised representation learning (Jaiswal et al., 2020). SSL approach allows the model to learn representations of the data in an unsupervised manner, using the relationships within the data itself as a guide. It can use a large amount of unlabeled data to pretrain encoders, thereby enhancing the performance of downstream main tasks. Unlike supervised learning, SSL does not require extensive manual labeling of data. Due to its ability to capture temporal dependencies and patterns embedded in unlabeled data, SSL has achieved strong performance in various time series-related tasks (Yang et al., 2025; Xi et al., 2022).

In the financial sector, SSL has various applications, such as learning financial signals, optimizing trading strategies, and predicting stock prices (Ma et al., 2022). Ying et al. (2024) proposed the SMART framework, which uses a SSL model to automatically learn hidden representations from raw stock time series data. Sang et al. (2025) further extended SSL in portfolio management by introducing a Multi-task Self-Supervised Learning (MTSSL) module that uses relational networks and auxiliary asset-comparison tasks to extract stable and generalizable state representations from noisy financial environments. Additionally, according to research by Yu (2018), high-dimensional features can affect the updates of Reinforcement Learning (RL) agents and decrease sample efficiency. Therefore, embedding input features into meaningful low-dimensional representations is crucial, as it can effectively learn useful features from large amounts of high-dimensional data, thereby improving model performance and efficiency. In summary, it is evident that SSL has a wide and important application prospect in various fields, including finance.

### 2.3. Contextual reinforcement learning

Contextual Reinforcement Learning (CRL) is a significant variant of reinforcement learning, allowing an agent to learn and make decisions within a given context. Here, the context can be considered any external factor that may impact the learning and actions of the agent, such as environmental status, time, previous experiences, etc. Compared to traditional reinforcement learning, CRL introduces a more flexible learning framework, enabling the agent to better adapt to dynamically changing environments and make optimal decisions.

The importance of CRL lies in its ability to allow the agent to learn and optimize different strategies in varying contexts. This parallels the way humans make decisions in the real world, as our choices are typically based on our specific situation and context. For instance, in financial markets, a trader's decisions might be influenced by market conditions, historical data, time, and other factors. CRL models can simulate this context-specific decision-making process, thus achieving more efficient and accurate decisions in various practical applications, such as financial trading, autonomous driving, etc. (Hallak et al., 2015) initially defined the Contextual Markov Decision Process (CMDP) that incorporates environmental contextual information into the traditional MDP process. Zong et al. (2024) applied CRL to high-frequency trading

by training specialized sub-agents under different market conditions, showing improved adaptability and performance.

For the option market-making problem, we consider market information such as underlying price changes, volatility, and investors' sentiment as market context. This means that the agent should consider not only high-frequency limit order book information but also the market context to ensure it can adapt to different market conditions.

### 2.4. Continual learning

Deep learning plays a crucial role in solving various complex problems, yet traditional deep learning models face a significant challenge: they typically learn one task at a time and may forget the knowledge learned from previous tasks when learning new ones. This phenomenon is also known as "catastrophic forgetting". The core concept of Continual Learning (CL) is to enable models to learn new tasks while retaining knowledge of previous tasks (Parisi et al., 2019; Han and Liu, 2022).

The importance of CL in deep learning is self-evident. It simulates how humans learn new knowledge sequentially and allows AI systems to better adapt to dynamic environments and continually changing tasks. Concepts and techniques of CL can be applied to various deep learning fields, including natural language processing, computer vision, and reinforcement learning. De Lange et al. (2021) categorizes the techniques of CL into three types for task incremental classification: replay methods, regularization-based methods, and parameter isolation methods. Building on this, Wang et al. (2024) presents a more comprehensive taxonomy in their survey, dividing CL methods into five major categories: regularization-based, replay-based, optimization-based, representation-based, and architecture-based approaches. In summary, CL is an important research direction in deep learning, particularly in reinforcement learning. It allows deep learning models to better adapt to dynamic environments, enhancing the practical value and application prospects of deep learning.

## 3. Methodology

In this section, we first outline the system architecture in Section 3.1, and we then explain the implementation details of SSMCN in Section 3.2. The design details and training process of CRL are introduced in Section 3.3. In Section 3.4, we elaborate on the method and process of CL.

### 3.1. System overview

Motivated by Wen (2023), we proposed a Contextual Reinforcement Learning framework for Market Making (MM-CRL), which is depicted in the system overview diagram in Fig. 1. The first section is the Self-Supervised Market Context Network (SSMCN). The goal is to elegantly transform the original high-dimensional and noisy financial market sentiment indicator series into a representative market context through a variety of auxiliary tasks. Next, the second section Contextual Reinforcement Learning (CRL) architecture uses the encoded market context outputted by SSMCN as an additional input alongside standard market features and the agent's internal trading state. It aims to train an agent that is environmentally aware and generalizes across multiple market-making environments. Furthermore, it integrates a curriculum learning method to cluster the training environments, ensuring the agent's convergence and training stability. The third section is the Continual Learning (CL) module which handles the deployment phase, where the agent will continuously face new market environments. By appropriately selecting training environments for fine-tuning the agent, it can keep the agent effective and alleviate the problem of concept drift. In the following sections, we will detail the operation of each block.

In summary, these three components are designed to operate in sequence: SSMCN learns an informative summary of the market condition, CRL uses that summary to help the agent adapt its policy across different market situations, and CL ensures long-term adaptability.

### 3.2. Self-supervised market context network

#### 3.2.1. Market context feature design

As outlined in Section 1, concerning recent market trends and sentiment is important for market-making tasks. We strive to gain insights into market dynamics from historical trends and shifts in sentiment indicators. However, financial data is usually full of noise (Giles et al., 2001; Wang and Wang, 2024), and utilizing high-dimensional time-series data may lead to the curse of dimensionality (Barto and Mahadevan, 2003). As such, it is essential to convert the data into a low-dimensional, representative latent vector. SSMCN proficiently consolidates similar market situations in vector space, which aids downstream market-making tasks by providing a context-based decision-making foundation. The input of SSMCN is a matrix denoted as $X_t^{mkt}$, which is a set of several time series related to the market with a daily frequency and a fixed lookback window size $lookback^{mkt}$. The components included are shown below:

- Daily price return of the TAIEX index
- VIX of TXO
- Put–call ratio of TXO
- Open interest of dealers (TXO & TAIEX Futures (TXF))
- Open interest of foreign investors (TXO & TXF)
- Open interest of top 10 retail investors (TXO & TXF)

According to the contents of the aforementioned list, $X_t^{mkt}$ includes both overall market indicators and various position indices of influential investors. The significance and relationship of each indicator with the market will be explained in the following paragraph.

The Taiwan Stock Exchange Capitalization Weighted Stock Index (TAIEX) price return is a critical measure of the overall performance of the Taiwan stock market. It reflects the price changes of all listed stocks on the Taiwan Stock Exchange (TWSE) and is weighted according to their market capitalization. In the options market, TAIEX price serves as an underlying price for index options. We use price returns instead of raw price data as input feature sequences, as price returns offer a standardized representation of market price fluctuations. For market making, this helps the agent distinguish bullish and bearish regimes, guiding whether to adopt conservative or aggressive quoting strategies. The calculation formula is as Eq. (1),

$$Return_t = \frac{P_t^{TAIEX}}{P_{t-1}^{TAIEX}} \tag{1}$$

where $P^{TAIEX}$ is the price of TAIEX.

The compilation method of the TXO's VIX is identical to CBOE's VIX, with the objective of calculating market investors' volatility expectations for the upcoming 30-day period in the stock market. This index serves as a reference for trading and hedging. A higher VIX index implies greater expected market volatility by investors, suggests widening spreads to reduce inventory risk, while a lower one indicates a more placid market, allows tighter spreads to capture more trades. The VIX is a commonly seen financial sentiment indicator that reflects the degree of change in the overall market investor's psyche, and is thus also referred to as the "Fear Index".

The put–call ratio is a significant indicator used in the options market, which provides insights into the market's sentiment. It is calculated by dividing the open interest of put options by the open interest of call options. A high ratio (above 1) is typically interpreted as a bearish sentiment, indicating that more investors are buying put options in anticipation of a price, which implies a more cautious quoting to avoid excessive long inventory. Conversely, a low ratio (below 1) suggests a bullish sentiment, where more traders are buying call options expecting the price to rise, justifying more aggressive bid quoting. Hence, the put–call ratio is a valuable tool for gauging market sentiment and forecasting potential shifts in market trends, as shown in Eq. (2), where

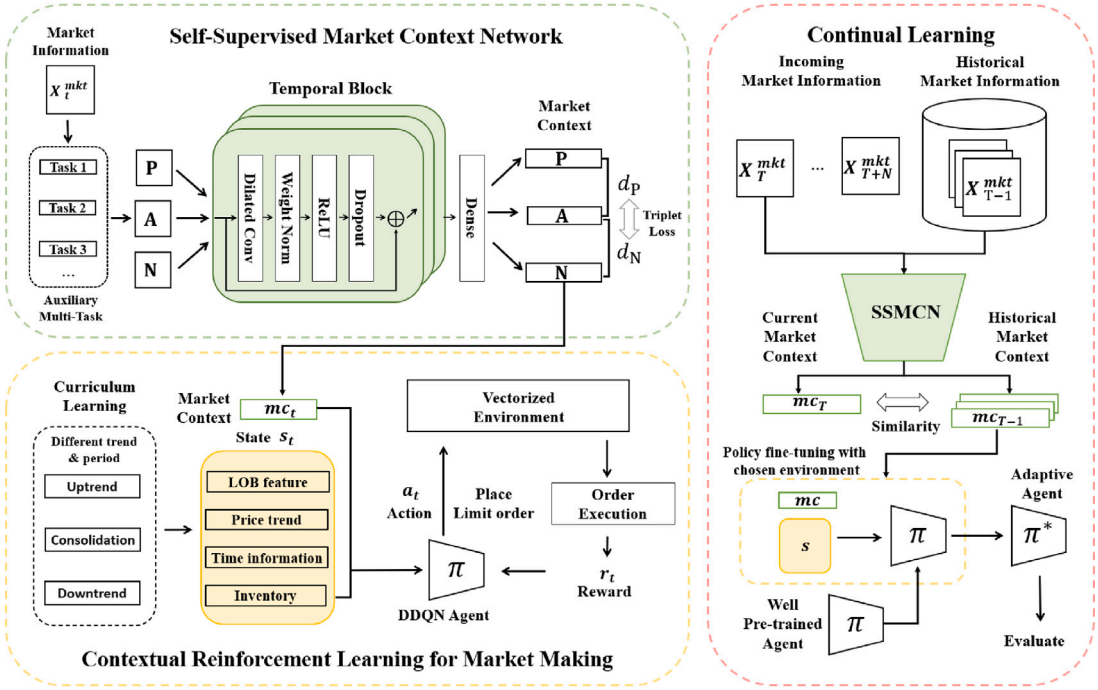


**Fig. 1.** System overview of MM-CRL.

*Put OI* and *Call OI* represent the open interest of put and call options, respectively.

$$\text{PC ratio} = \frac{Put\ OI}{Call\ OI} \tag{2}$$

Open interest refers to the total number of outstanding derivative contract that have not been settled. In the options market, it is an important indicator that reflects the activity level of a specific contract, changes in open interest can also provide insights into market sentiment. We use the open interest data of three investor roles, namely foreign investors, proprietary traders, and retail investors (based on the top ten open interest volumes). Each of these principal actors has different market expectations and strategies. Including these data points can help the model incorporate more diverse perspectives and enhance its predictive ability regarding future market trends. For options, as there are both call and put, and buyers and sellers, therefore end up with four sequences in total. For futures, there is only two sequences, buyers and sellers, therefore the input features for each investor will consist of six sequences. The reason we incorporate futures data is that it offers the opportunity to observe the hedging direction of proprietary traders and provides more insights into market investors' expectations of Taiwanese stocks.

### 3.2.2. Model architecture and loss function

In terms of self-supervised methods, we have chosen the Temporal Convolutional Network (TCN) (Lea et al., 2017) as our model, and we update it using the triplet loss function. TCN offer several advantages when extracting features from financial time series data, such as learning hierarchical features in an end-to-end manner, which can reveal complex, non-linear patterns in financial markets. Also, it can capture long-term dependencies in time series data effectively, which is critical for understanding market trends. Prior work has also demonstrated the effectiveness of TCNs specifically for financial time series modeling (Sun et al., 2022b), supporting our architectural choice. Triplet loss is a type of loss function used in machine learning to train models to learn meaningful representations of data. It operates on triplets - an anchor data point, a positive point (similar to the anchor), and a negative point (dissimilar to the anchor). The objective is to make the anchor point closer to the positive point than the negative point in the feature space, effectively learning the relative similarities and differences between data points. This method is capable of learning significant representations of financial data, capturing the subtle relative similarities and differences. By learning the feature representation of financial data in this manner, the model can make more accurate predictions by identifying key trends and patterns in the raw data that are not readily apparent. The formula for the triplet loss $L$ is shown in Eq. (3),

$$L = \max\left(\|f(A) - f(P)\|_2^2 - \|f(A) - f(N)\|_2^2 + \alpha, 0\right) \tag{3}$$

where: $f(A)$ represents the embedding vector of the anchor point, $f(P)$ is the embedding vector of the positive sample, and $f(N)$ is the embedding vector of the negative sample. $\alpha$ is the margin, which ensures that there is at least a certain gap between the distances of the positive and negative samples. $f$ is the encoding function, and in our framework, we use the TCN as $f$.

### 3.2.3. Auxiliary task design

In our research, we strategically devised different auxiliary tasks intended to optimize our market context in order to more accurately reflect the daily market conditions. Specifically, we used the financial sentiment indicators from the next day as our learning target. In these auxiliary tasks, we categorized scenarios with similar indicator values as positive samples, and instances with more distant values as negative samples. Our objective was to bring representations of similar market scenarios closer in the vector space and distance the representations of dissimilar ones, enhancing its ability to distinguish different market scenarios.

Below, we provide detailed explanations regarding the significance of these auxiliary tasks and the reasoning behind our choice of these tasks. We anticipate that these designs will aid our agent in better learning and understanding market scenarios, subsequently improving its performance in downstream market making tasks. Our auxiliary task uses various financial indicators from the next day as labels as follows:

- TAIEX price return from the next day
- VIX from the next day
- Put-Call ratio from the next day

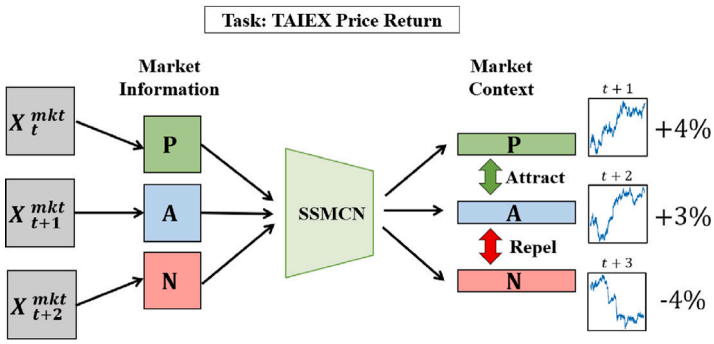


**Fig. 2.** Illustration of SSMCN task.

- Long-Short ratio of investors from the next day

We have initially designed three auxiliary tasks closely related to overall market dynamics: the TAIEX price return, the VIX futures index, and the put–call ratio. These tasks respectively represent the market trend, market volatility, and investor sentiment, which we believe are crucial indicators for market-making.

The TAIEX price return task aims to enable the model to learn that under similar market situations in terms of the next day's price movement, their market contexts should be similar. Through this task, we hope that the model can better judge today's market trend and adjust its order placing strategy accordingly, such as leaning towards long or short positions. An illustration of this task is shown in Fig. 2, where the objective of the SSMCN is to bring the market contexts of similar future trends closer together. The VIX futures index is related to the expected market volatility. Our goal is for the model to consider market contexts similar when the expected market volatility is similar. The volatility index can help the model predict the potential range of market fluctuations. When the expected market volatility is higher, the market may be more unstable, and the model might need to adjust its order placing strategy, such as widening the order spread. The put–call ratio represents the bullish or bearish sentiment of investors in the options market. We hope the model can learn that when the overall investor sentiment for the next day is similar, their market contexts should be similar. A high put–call ratio might indicate bearish sentiment in the market, but it could also result from the situation where put option holders are stuck in a bullish market, leading to a greater put option position than call options.

Next, we designed six auxiliary tasks targeting three major market participants: foreign investors, proprietary traders, and retail investors. These tasks involve calculating the long-short ratios of their option and futures positions. We believe that the long-short ratio can reflect investor sentiment and provide useful market information, which is because these market participants are the primary drivers of market prices, and their trading decisions have a profound impact on market trends. Therefore, we use the holding situation of these investors as the goal of our auxiliary tasks to help the model understand changes in market sentiment. Furthermore, the trading behaviors of these participants also serve as references for most retail investors and influence their trading decisions. This reasoning is an important factor in why we choose these auxiliary tasks. By learning from these tasks, our model can gain a comprehensive understanding of the dynamics of various market participants, thereby enabling it to make better trading decisions in different market contexts.

These auxiliary tasks also play a crucial role in enhancing overall performance. They enhance feature extraction by encouraging the model to capture important patterns beyond those directly tied to the reward, improve generalization by providing predictive signals that remain informative across different market conditions, and strengthen risk awareness through tasks such as trend prediction and volatility estimation. Through joint optimization with the main reinforcement learning objective, the auxiliary tasks ensure that the downstream market-making policy is more adaptive, robust, and better aligned with real-world trading conditions.

Where the long-short ratio represents the bullish or bearish sentiment expressed by an investor's open interest volume. This ratio is calculated separately for options and futures. The long-short ratio of options, as shown in Eq. (4), is derived by dividing the open interest of bullish positions by the bearish positions, where $OI_{call}^{long}$ denotes the open interest of long call positions, $OI_{put}^{short}$ denotes the open interest of short put positions, $OI_{call}^{short}$ denotes the open interest of short call positions, and $OI_{put}^{long}$ denotes the open interest of long put positions. If the ratio is greater than 1, the sentiment leans towards bullish; if it is less than 1, it indicates a bearish sentiment.

$$LS_{Options} = \frac{OI_{call}^{long} + OI_{put}^{short}}{OI_{call}^{short} + OI_{put}^{long}} \tag{4}$$

For futures positions, the calculation involves subtracting the sell-side open interest (OI) from the buy-side OI, and then dividing the result by the total OI. A positive number indicates a predominantly long position, while a negative number indicates a short position. The calculation formula is presented in Eq. (5).

$$LS_{Futures} = \frac{OI^{long} - OI^{short}}{OI^{Total}} \tag{5}$$

It should be noted that the long-short indicator merely reveals the direction of open interest and does not necessarily represent an investor's absolute bullish or bearish outlook. In fact, the interpretation can vary depending on the market conditions and the use of different indicators. In our tasks, we separately calculate the long-short ratio for foreign investors, proprietary traders, and retail investors in both options and futures, thus generating a total of six tasks.

However, it is crucial to point out that the market sentiment expressed by the put–call ratio and the long-short ratio indicators merely indicates a bias in open market positions, and does not definitively represent investors' bullish or bearish attitudes. In reality, market participants may make varied trading decisions based on the current market conditions and other financial indicators. Thus, we must incorporate information from multiple factors into our considerations, bearing in mind that the relationships among these factors are highly complex. This is a key reason for our adoption of a deep self-supervised network for feature learning. We anticipate that through SSMCN (Self-Supervised Market Context Network), we can extract useful market context from complex market information. By integrating these contexts, we aim to enhance downstream agents' capacity to better understand and adapt to different market environments.

### 3.2.4. Training procedure

The algorithm of SSMCN training process is presented in Algorithm 1. First, we create independent datasets for each auxiliary task, which includes multiple windows of $X^{mkt}$ and the corresponding label (i.e., the sentiment indicator for the following day), the window size

is $lookback^{mkt}$. For example, $X_t^{mkt}$ includes all financial sentiment indicators from day $t - lookback^{mkt}$ to day $t$, while the label represents the indicator for the day $t+1$ (such as price return or VIX). During training, we extract triplets of the batch size. In each triplet, the first sample is always the anchor, while the remaining two samples are determined as positive or negative based on the distance of their labels. All auxiliary tasks share the TCN backbone, which consists of five temporal blocks with dilated 1D convolutions, residual connections, ReLU activations, and dropout layers. The training process is completed when all tasks have undergone multiple rounds of training until the model converges.

**Algorithm 1** SSMCN Training Process

**Input:** Market Information, $X^{mkt}$, Auxiliary Tasks , $T$; Datasets, $D$; The initial parameter of the SSMCN, $\theta_{SSMCN}$;
**Output:** Market Context, $mc$ ;

```
for number of training episodes do
    for task t in T do
        Set dataset d from D
        Sample batchsize Triplets from d
        Set positive & negative sample
        Update θ_SSMCN via triplet loss
    end for
end for
mc ← f_{θ_SSMCN}(X^mkt)
return mc
```

### 3.3. Contextual reinforcement learning

In this section, we will first introduce the problem definition, market making procedure and limitations of our framework, follow with introducing state, action, and reward used in contextual reinforcement learning. Finally, we introduce the curriculum learning mechanism to stabilize our training process.

#### 3.3.1. Problem formulation

We regard the market making problem as a complex construct, specifically a Contextual Partially Observable Markov Decision Process (CPOMDP). This framework combines the strengths of both Contextual Markov Decision Process (CMDP) and the Partially Observable Markov Decision Process (POMDP). The CMDP (Hallak et al., 2015) takes into account the influence of environmental context, providing additional information about the environment so that an agent can make better decisions based on the specific context, which allows our agent to adaptively adjust strategies across bullish, bearish, or sideways markets instead of treating them as a single homogeneous environment. The POMDP (Kaelbling et al., 1998) considers the situation of incomplete observation. In POMDP, an agent cannot directly observe the complete environmental state, and must infer the market state and make decisions based on partial observations. As in real trading, the agent never fully observes the true market state since many factors (e.g., hidden order flow, news shocks) are unobservable. Thus, it must rely on partial observations such as the limit order book and contextual embeddings to infer the latent state.

Formally, we define the CPOMDP as $\langle S, A, P, R, \gamma, O, \Omega, C\rangle$, where $S$ is the state space, $A$ is the action space, $P$ is transition probability, $R$ is the reward function and a discount factor $\gamma$. In trading problems, the agent cannot fully observe the state of the market environment, as market prices may be influenced by many factors, such as news events, behaviors of other traders, and so on. Therefore receives an partially observation $o$ from $O$, with observation probability $\Omega(o|s,a)$. Finally, there is the context space $C$. The context is related to the transition process of the POMDP, so the probability transition function is denoted as $P(s'|s,a,c)$. In our implementation, the SSMCN encodes historical sentiment into a context vector C, which is concatenated into the agent's input state. This design allows the agent's policy to condition decisions on market context, effectively reflecting the CPOMDP formulation without explicitly modifying the environment transition.

In our formulation, each trading day operates based on a fixed but unobserved market context, and the sequence of market making interactions within that day (from open to close) is modeled as a POMDP. The goal of the agent is to learn to adjust its market making strategy for the day according to the market context, thus enhancing its generalization capabilities.

#### 3.3.2. Market making procedure

In our problem setting, the market making process does not operate on a fixed-time interval, but rather place orders in response to changes in the LOB, for instance, according to Fig. 3, the agent requotes when there are 20 LOB changes. The interval of 20 order book updates was selected as a hyperparameter through parameter exploration. In our preliminary experiments, we tested a range of interval settings and observed that shorter intervals resulted in minimal price movement and sparse rewards, while longer intervals reduced the likelihood of timely execution. Thus, 20 updates emerged as a suitable balance, providing sufficient price variability for effective learning while maintaining execution feasibility. We believe that the agent should adjust its order placement according to the level of market activity instead of a fixed time interval. If a fixed time interval approach is adopted (e.q. every 5 s), it is likely that when during periods of low market activity, there will be almost no trading activity, leading to reward sparsity and instability.

The market making process, as shown in Fig. 3, begins with the agent observing the state, which typically consists of key market information such as the best bid and ask prices and sizes from the order book, the agent's current inventory position, and recent trading activity. Based on these observations, the agent selects an action, which determines how many ticks away from the best bid–ask price to place the order. The agent then waits for a period for the order to be executed (20 LOB changes as the example shown in the figure). The execution of the limit order is determined when the best bid or ask price crosses the order price. We assume that the order is at the end of the order queue and disregard the quantity of the order (set as one lot). Regardless of whether the order is executed or not, a reward is calculated, completing one cycle until the next order is placed. In our framework, the agent's state includes not only standard information but also a richer set of technical and contextual indicators extracted through SSMCN. while the action determines the quoting distance in ticks from the best bid and ask. Detailed definitions are provided in Sections 3.3.4 and 3.3.5.

#### 3.3.3. Limitations

The market making problem in this study includes the following assumptions and limitations:

1. We assume that the market maker's orders are always placed at the end of the order book. A trade is executed once the market price crosses the limit order price.
2. The quantity of executed trades is not considered, each trade is executed with a fixed size of one lot.
3. No transaction costs are included in the trades. That is, no additional fees or taxes are deducted for each transaction.

In typical order-driven financial markets, trades are executed according to a price-time priority rule, where orders at better prices are matched first, and among orders at the same price, those submitted earlier take precedence. This assumption follows the trading rules adopted by most exchanges, thereby reflecting real-world market practices. To avoid slippage and matching priority simulation error, we adopt a stricter implementation in this study. Each limit order submitted by the agent is placed at the end of the current order queue at the specified price level. During the order's lifetime – i.e., until the agent updates or cancels the order – a trade is considered executed only if the market price crosses the limit price. In reality, however, execution can occur even without a full price crossing. By requiring such a crossing to

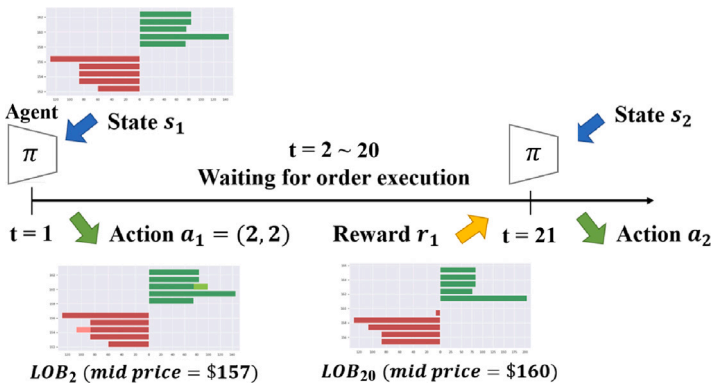


**Fig. 3.** Market making procedure.

generate profits, our design remains consistent with what can occur in real markets.

Market makers of derivative financial products in Taiwan are required to obtain approval from the competent authority and are eligible for transaction and clearing fee incentives offered by the exchange. Equivalently, those who fulfill all obligations and attain the specified performance benchmarks may receive up to a 100% reduction in trading costs. In addition to fee reductions, market makers may also benefit from revenue-sharing arrangements, relaxed trading restrictions, and exclusive product distribution rights, thereby gaining additional profitability and competitive advantages. As in previous literature (Falces Marin et al., 2022), the market-making strategy can therefore be reasonably modeled under a zero transaction cost assumption. However, since our experimental setting is based on the Taiwanese market, where fee incentives are provided, transaction costs were excluded. In markets without such incentives, this exclusion could lead to an overestimation of profitability, particularly for high-frequency strategies where costs accumulate rapidly. Therefore, the current setting may not be directly applicable to other countries or exchanges with different trading mechanisms.

Moreover, in practice, noisy input data (e.g., due to erroneous feeds or latency), distributional shifts between training and live environments (e.g., changing market dynamics), and the use of sentiment indicators (e.g., VIX, open interest) as auxiliary signals may not generalize well across all markets due to potential data unavailability or inconsistencies. Moreover, applying such methods to other domains (e.g. manufacturing) can introduce additional challenges, as industry-specific dynamics, data structures, and response behaviors (e.g., supply chain shocks or physical constraints) may differ significantly from financial contexts. Nevertheless, despite these limitations, we still offer insights into how to use various advanced deep learning techniques to construct more intelligent market making strategies.

### 3.3.4. State

The state includes various information that might be used in high-frequency market making tasks, including order book information, options pricing indicators, options product information, and inventory holdings. Each of these elements will be detailed below, and the numbers in parentheses represent the lengths of the feature vectors.

- **Order book imbalance**: It is obtained by subtracting the sum of the top five bid volumes from the sum of the top five ask volumes, and then dividing the result by the total volume. It can be used to measure short-term buying and selling pressure, provides practical implications for market making: a positive value (buy-side depth > sell-side depth) indicates higher probability of bid orders being executed and thus greater risk of accumulating long inventory, whereas a negative value implies stronger sell pressure. As shown in Eq. (6),

$$\text{Orderbook Imbalance} = \frac{\sum_{i=1}^{5} V_i^a - \sum_{i=1}^{5} V_i^b}{\sum_{i=1}^{5} V_i^a + \sum_{i=1}^{5} V_i^b} \tag{6}$$

where $V_i^a$ and $V_i^b$ are ask and bid volume of different price level. This feature retains a sequence over a certain past period, with a length of $lookback^{state}$.

- **Mid price return**: Represents the mid-price change in high frequency, where the mid-price is the average of the best bid and best ask prices, as Eq. (7),

$$\text{Mid Price} = \frac{(P_0^a + P_0^b)}{2} \tag{7}$$

where $P_0^a$ denotes best ask price and $P_0^b$ denotes best bid price. This price series is converted into a return rate for standardization and to denote changes in price increases or decreases. This series can be regarded together with the order book imbalance to understand short-term transaction information and changes in buying and selling pressure. This feature retains a sequence over a certain past period, with a length of $lookback^{state}$.

- **Minute-level price return**: To represent longer-term price trends, we include the opening, high, low, and closing prices from the previous 5 min as additional information. All prices are normalized by dividing them by the current mid-price. If there are no transaction data available for a given minute, we use the price from the previous minute to impute missing values. This feature retains a sequence over a certain past period, with a length of $lookback^{state}$.
- **Time to market close**: We believe understanding the market's current moment is also crucial. For instance, the market is typically more active at the opening and near the closing time. Understanding the time left until market close can help an agent adjust its market making strategy, such as its spread, and potentially learn about some cyclical changes. This is calculated as Eq. (8),

$$\text{Time To Market Close} = \frac{(T_{close} - T_{now})}{T_{total}} \tag{8}$$

where $T_{total}$ is the total market opening time, $T_{now}$ is the current time and $T_{close}$ is the time when market closed. Time is measured in seconds, and the value of time to market close will fall between 0 and 1, where a value closer to 0 indicates a time closer to market close.
- **Implied volatility**: The implied volatility is an important indicator of current investors' expectations about market volatility and is also one of the factors influencing option prices. We calculate implied volatility using a bisection method. Given a target market price for an option, we evaluate the Black–Scholes (Black and

**Table 2**
Action space for market making.

| Action ID | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| Ask $\delta_a$ | 0 | +1 | +2 | +3 | 0 | +2 | +1 | +3 |
| Bid $\delta_b$ | 0 | −1 | −2 | −3 | −2 | 0 | −3 | −1 |

Scholes, 1973) formula at the midpoint and iteratively narrow the range in which the implied volatility may exist. This process repeats until the estimated price is sufficiently close to the market price. The Black–Scholes pricing formula used for call options are as Eq. (9),

$$C(S_t, t) = S_t N(d1) - Xe^{-r(T-t)} N(d2) \tag{9}$$

where $S$ represents the price of the underlying asset, $X$ is the strike price, $T$ is the time of settlement, $t$ is the current time, $N()$ is the cumulative distribution function, and $r$ is the risk-free interest rate, which is derived from the yield of a ten-year Taiwanese government bond. Also $d1 = \frac{1}{\sigma\sqrt{T-t}}\left[ln\left(\frac{S_t}{X}\right) + \left(r + \frac{\sigma^2}{2}\right)(T-t)\right]$, $d2 = d1 - \sigma\sqrt{T-t}$, and $\sigma$ represents the volatility we seek. While knowing the option price and other factors, we can infer the implied volatility using the bisection method.

- **Log moneyness**: To express the degree to which an option is in-the-money or out-of-the-money, if the price gradually leans towards being deeply in-the-money or deeply out-of-the-money, the trading of the product may gradually cool down. In addition, this information is also one of the parameters for option pricing. The calculation formula is as Eq. (10),

$$\text{Log Moneyness} = log(\frac{P_t^{TAIEX}}{K}) \tag{10}$$

where $P_t^{TAIEX}$ is current price of underlying index and $K$ is strike price of the corresponding option.

- **Time to maturity**: Options that are closer to their expiration date have lower time values. Apart from affecting market prices, this characteristic can also be used to learn the relationship between the time to expiration and market trading behavior. As we have selected option contracts expiring within about a month, we normalize using 30 days as the denominator.
- **Inventory**: The current inventory position has a decisive impact on the agent's adjustment of its order placement strategy. The inventory position is measured in lots, with a inventory limit constraint. In our setup, it is possible to take short positions in contracts, allowing the inventory position to be negative. The formula for standardizing the inventory position using the inventory limit to constrain the values within the range of $[-1, 1]$. The equation is given by Eq. (11),

$$\text{Inventory} = \frac{Inv_t}{Inv_{limit}} \tag{11}$$

where $Inv_t$ is current inventory position and $Inv_{limit}$ is the inventory limit.

### 3.3.5. Action

The action space is a finite discrete set $A = \{a_0, a_1, a_2..., a_7\}$, where each action $a = (\delta_a, \delta_b)$ represents the ask-bid spread pair, where $\delta_a$ represents how many ticks away from the best ask price, while $\delta_b$ represents how many ticks away from the best bid price. When the agent executes an order placement, the order placement points $\delta_a$ and $\delta_b$ are multiplied by the tick size $\Delta_{tick}$ and then converted into the order buying price and selling price, as shown in Eq. (12).

$$\begin{aligned} P^{ask} &= \delta_a * \Delta_{tick} + P_0^a \\ P^{bid} &= \delta_b * \Delta_{tick} + P_0^b \end{aligned} \tag{12}$$

The spread of action is shown in Table 2, where we have chosen only eight representative actions with the intention of preventing an overly large action space, which could impede the convergence of the model. Please note that in the option products we use, the tick size $\Delta_{tick}$ changes according to the options' price. The correspondence table is shown in Table 3.

**Table 3**
Tick size of each price range.

| Price range | Tick size $\Delta_{tick}$ |
|---|---|
| $P < 10$ | 0.1 |
| $10 \le P < 50$ | 0.5 |
| $50 \le P < 500$ | 1 |
| $500 \le P < 1000$ | 5 |
| $1000 \ge P$ | 10 |

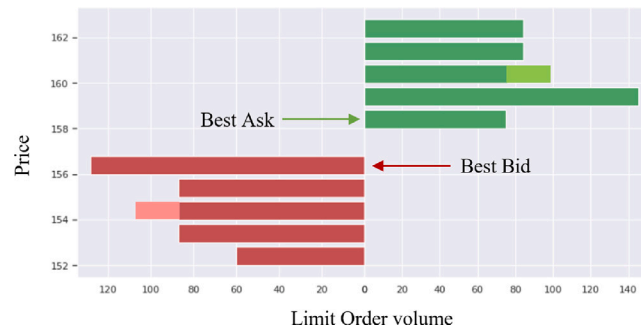


**Fig. 4.** Limit order book with action.

Here is an example of order placement, if the current best bid price is \$156, best ask price is \$158, and the tick size is equal to \$1. When action is $a_2 = (+2, -2)$, the market maker will quote at \$154 and \$160, as shown in Fig. 4. In another case, if action is $a_6 = (+1, -3)$, the quote would be \$153 and \$159. Smaller spread means more aggressive to the market, i.e. the order would be easier to be filled, but with less profit of price spread. Skew of the spread is used as controlling inventory risk, if the market maker has a large long position currently, the quote is preferred to be $a_4$ or $a_6$, which means it is more possible to sell assets than buy assets, in order to reduce the inventory. In this way, each action corresponds to a realistic market-making adjustment: narrowing the spread increases trade frequency but raises inventory risk, widening the spread lowers execution probability but mitigates inventory accumulation, and skewing quotes towards one side directly manages exposure. This design reflects the fundamental trade-offs of market making between liquidity provision, spread capture, and inventory control. Finally, as mentioned in Section 3.3.2, our model is based on an assumption: as long as the market price exceeds our order price, the order can be traded. Under this assumption, the quantity of the order is not a major consideration in our problem setting. Therefore, we uniformly use a quantity of 1 lot as the trading unit when placing orders.

### 3.3.6. Reward

In the problem of market-making, the primary objective is to maximize the profitability of the strategy while ensuring that the inventory position does not accumulate excessively. There are two sources of profit. The first type is realized through the bid–ask spread. When a market maker place sell orders above the mid-price and buy orders below it, if both orders are executed, the bid–ask spread is earned. This type of profit does not cause changes in inventory positions and is the ideal way for market makers to profit. The second type is profiting from an increase in the value of positions due to market price changes. As the inventory position increases, market makers have the opportunity to earn large returns, but this is actually a high-risk move and could easily lead to a loss due to rapid price changes. A market-neutral and conservative market maker would choose to maintain as

small an inventory position as possible, earning the spread through continuous two-sided trading. Our reward formula is as Eq. (14), where $\phi_t$ represents the profit from the bid–ask spread, $q_t$ represents the inventory, $p_t^{mid}$ represents the current mid price, and $\Delta m_t$ represents the mid price change. To account for differences in price scales across option products, the reward is normalized by $\Delta tick$, so it reflects profit and loss in units of ticks rather than absolute price values. This design aligns with the Taiwan index options market, where tick sizes are fixed by exchange rules based on price levels and do not fluctuate over time.

$$\begin{aligned} \phi_t^a &= p^{ask} - p_t^{mid} \text{ if matched else } 0 \\ \phi_t^b &= p_t^{mid} - p^{bid} \text{ if matched else } 0 \\ \phi_t &= \phi_t^a + \phi_t^b \end{aligned} \tag{13}$$

$$R(t) = \frac{(\phi_t + q_t \Delta m_t)}{\Delta_{tick}} - \eta \|q_t\| \tag{14}$$

To prevent the agent from holding a high inventory position, we include an inventory penalty $\eta\|q_t\|$, where $\eta$ is a hyperparameter. The higher the inventory position, the greater the negative reward incurred, thus compelling the agent to actively reduce its position. This reward design reflects the practical need in market making to balance short-term profitability with long-term inventory risk, where the penalty term acts as a soft constraint to discourage excessive inventory accumulation without imposing hard limits. To empirically validate the effectiveness of this reward design, we further analyzed the impact of different penalty coefficients on trading performance, which directly corresponds to RQ1. Specifically, we conducted ablation and sensitivity experiments by varying the value of $\eta$. The results confirmed the expected trade-off: smaller penalty terms encouraged higher profitability but increased inventory exposure, whereas larger penalties effectively constrained inventory risk at the cost of reduced profit. These findings demonstrate that the proposed reward structure successfully captures the essential balance between profitability and inventory control in market making.

#### 3.3.7. Curriculum learning

Given the diversity of the training environment, it is challenging to have the agent interact with all environments and update at once. During the initial stages, the agent may struggle to collect regular experiences, leading to difficulties in achieving convergence during training. Therefore, we draw from the concept of curriculum learning (Narvekar et al., 2020), reducing the difficulty of individual tasks and enabling the agent to make progressive updates by categorizing tasks. In curriculum learning, the model is initially trained on simpler tasks and gradually progresses to more complex ones. This strategy allows the model to better internalize the underlying distribution of the training data and reduces the risk of overfitting to complex tasks during the early stages of training. Additionally, this learning strategy enhances the efficiency of the model's learning process, since it establishes a foundation of knowledge on simpler tasks before transitioning to more complex ones.

To apply the strategy to our problems, we categorize the training environments into six groups according to the year and the strength of price trends. The strength of a trend is calculated by dividing each day's closing mid-price by the opening mid-price, and all the market environments for that year are ranked. Environments with values less than the first quartile (Q1) are considered downtrend environments, those between Q1 and the third quartile (Q3) represent sideways trends, and those exceeding Q3 indicate uptrend environments. This categorization reflects different levels of market complexity: trending markets provide clearer directional signals that simplify learning, whereas sideways markets are more ambiguous and difficult, making them suitable for later stages of training. We conduct training sequentially following the order of uptrend, downtrend, and sideways trend for 2017 and then 2018. Each training session interacts with the environment in parallel using a vectorized environment, which speeds up the training process. This staged progression is designed to stabilize early learning, prevent premature convergence, and allow the agent to first master basic strategies before tackling more volatile or complex environments. By gradually increasing task difficulty, the curriculum enables the agent to acquire robust trading policies and enhances overall training effectiveness.

### 3.4. Continual learning

In the context of CL mechanisms, our objective is to select historical environments that resemble the current market situation, enabling the agent to appropriately fine-tune its policy. We use the market context generated by SSMCN to represent the day's market situation, and decide which historical environments are the most similar by calculating the market similarity between market contexts. The similarity is measured based on the Euclidean distance between vectors. Euclidean distance provides an intuitive way to quantify similarity between market contexts in the latent space, allowing structurally comparable states, such as those with similar volatility or order-flow patterns, to be grouped together for policy fine-tuning. This enables the agent to leverage prior experience from analogous environments and improves the stability of the fine-tuning process. Nevertheless, this approach also has limitations: it assumes equal weighting of all features after normalization, and may fail to capture nonlinear dependencies among market variables. These factors suggest that while Euclidean distance is effective as a baseline similarity measure, more sophisticated alternatives could be explored in future work. The updating mechanism is executed once a day before the market opens, to ensure that the model remains in its optimal status during the day's trading. We always use the pre-trained model from the training stage as the basis for updates. The reason for not using the model from the previous day for further updates is that the market trends for two consecutive days could differ significantly, which could make the adjustment process for the model from the previous day to suit the current market conditions more challenging. This approach allows us to focus only on daily fine-tuning, without worrying about cumulative errors from updates. The detailed algorithm is as Algorithm 2.

**Algorithm 2** Continual learning training process

**Input:** Pre-trained policy, $\pi$; Historical market context, $mc_{old}$ ; Current market information, $X_t^{mkt}$; Current market context, $mc_t$ ; Number of training environments, $N_{env}$; trained parameter of SSMCN, $\theta^*_{SSMCN}$;
**Output:** Updated policy, $\pi^*$ ;

```
for Days in testing period do
    mc_t ← f_{θ*_SSMCN}(X_t^mkt)
    for mc in mc_old do
        Calculate distance between mc and mc_t
    end for
    Sort distances and select top N_env environments
    Update π using selected environments
    π* ← π
    return π*
end for
```

We expect that the design of CL can enable the agent to continuously learn knowledge corresponding to the current market situation, to avoid the agent becoming ineffective due to the concept drift in financial market environments. Furthermore, we believe that CL is an essential approach for intelligent agents applied to financial trading, ensuring that strategies can continue to learn and persist in the market.

## 4. Experiments

In this section, we initially introduce the real-world dataset, evaluation metrics and comparative methods in Section 4.1. To substantiate the effectiveness of our approach and to analyze the experimental results, we present four research questions which are addressed individually following Section 4.3. Here are our research questions:

- **RQ1:** How to set the optimal inventory penalty hyperparameter.
- **RQ2:** Does the addition of different auxiliary tasks in the SSMCN contribute to performance improvement?
- **RQ3:** Does the use of market context to select environments for policy fine-tuning improve performance in test environments?
- **RQ4:** Does our method outperform other control groups?
- **RQ5:** Performance evaluation on an additional dataset.

Overall, the design of our experiments follows a structured and progressive logic. We begin with RQ1 to tune key hyperparameters and ensure the experimental setup is stable. Next, RQ2 and RQ3 serve as ablation studies, investigating the contribution of individual components and improvements within our proposed method. RQ4 focuses on comparing the full method against a range of external baselines to assess overall effectiveness. Finally, RQ5 evaluates the generalization ability of our proposed method by applying it to an additional dataset.

### 4.1. Dataset and experimental settings

We used tick data from the TAIEX Option (TXO), listed on the Taiwan Futures Exchange (TAIFEX). TXO is the most actively traded product in the Taiwan options market, and the underlying price of the options contract is the Taiwan Weighted Stock Price Index (TAIEX). Our dataset covers the period from January 2017 to June 2019, with the first two years used for training data and the last six months for testing. The options contracts we selected are due to expire within a month. Due to increased volatility and uncertainty on the delivery day, we excluded the data from that day, resulting in a total of 469 training days and 109 testing days.

Daily data includes tick-level limit order book information from the opening to the closing of the market, containing the best five buy and sell prices and the corresponding transaction volumes. Data is recorded every time the order book changes, resulting in irregular time intervals between data points. We noticed that the order book often displays unusually wide bid–ask spreads and irregular price fluctuations, introducing excessive noise into the data during market opening and closing, so we removed the abnormal data for the first and last five minutes of each day. In quantitative trading research, it is common practice to exclude the very beginning and the last few minutes of each trading day because these periods often exhibit irregular price fluctuations that introduce noise. As a result, most studies sacrifice trading opportunities during these intervals, and this preprocessing step should be regarded as a limitation of such methods. The proportion of data discarded by this preprocessing step is minimal, accounting for less than 1% of the total dataset. Our final training dataset is composed of data from 8:50 am to 1:40 pm each day. We selected the two most actively traded products (those with the most limit order book changes) every day, resulting in a total of 938 training environments and 218 environments for testing, with an average of approximately 31,171 limit order books data per day.

We present the parameters and additional configurations used during training. Firstly, we employed elegantrl (Liu et al., 2021) as our RL core. Elegantrl offers a lightweight training framework coupled with comprehensive evaluation functions, making it well-suited for academic research. We utilized the Double Deep Q-Network (DDQN) (Van Hasselt et al., 2016) as our policy, which is a stable and effective critic-based method for discrete action spaces. Table 4 shows the hyperparameters used in our experimental setting. The hyperparameter settings were determined based on a combination of manual tuning and small-scale grid search, particularly for learning rate and batch size, to achieve stable convergence within the available computational resources. During the process, we also incorporated domain knowledge, such as referencing parameter choices from prior work on related topics, to guide the configuration process. For instance, the window size used for capturing market dynamics was set to one month, a practical and commonly adopted period in financial literature. Likewise, the embedding dimensions were chosen based on standard $2^n$ sizes frequently used in representation learning research.

**Table 4**
Hyperparameters of the experimental setting.

| Framework | Description | Value |
|---|---|---|
| SSMCN | window size of market information $lookback_{mkt}$ | 30 |
| | market context dimension | 32 |
| | number of temporal block | 5 |
| | dropout ratio | 0.2 |
| | batch size of SSMCN | 1024 |
| CRL | window size of state $lookback_{state}$ | 20 |
| | layer of DDQN | 5 |
| | hidden layer dimemsion od DDQN | 512 |
| | batch size of DDQN | 16 384 |
| | buffer size | $2^{22}$ |
| | training step | 1000 |
| | reward discount factor $\gamma$ | 0.9 |
| CL | Number of training environments $N_{env}$ | 20 |

### 4.2. Evaluation metrics and baselines

The performance evaluation metrics we utilized in this experiment encompass a variety of profitability and risk indicators. These metrics effectively demonstrate the earning capability and stability of the strategy are shown as follows:

- **Profit and loss (PnL)**: The profit or loss value obtained at the end of each day's settlement, denoted in TAIEX points. This measure is used to demonstrate the profitability of the strategy, with a higher average value being better.
- **Mean Average Position (MAP)** (Spooner et al., 2018): The average size of the inventory position held during the daily trading process (measured in lots). In market making, a larger inventory position is seen as a greater risk, so a lower MAP is better. This implies that the strategy faces less inventory risk during market making.
- **Sharpe Ratio (SR)** (Sharpe, 1965): Profit divided by the standard deviation, used to measure the stability of the profit. A higher value is better.
- **PnL-MAP** (Gasperov and Kostanjcar, 2021): The PnL divided by the MAP, considered as a profit measure after risk standardization. A higher value is better, implying that the strategy has a higher profitability capability while taking on the same inventory risk.
- **Maximum Drawdown (MDD)** (Magdon-Ismail and Atiya, 2004): The maximum PnL decline experienced during daily market making strategy. A lower value is better, indicating more stable profit-making ability and the ability to avoid larger losses.
- **Final Inventory (INV)**: Refers to the average inventory held at the close of each trading day. A lower value is preferable, as it implies that market makers can reduce hedging costs and avoid overnight risk.

In selecting these metrics, we aimed to balance standard comparability and the unique features of market making. PnL and Sharpe Ratio are well-established benchmarks for profitability and risk-adjusted return, facilitating comparison with prior works. However, since market making involves continuous inventory management, we additionally adopt MAP and Final Inventory to capture position exposure. PnL-MAP further integrates profitability with inventory risk, making it particularly suited for this context. Unlike general trading strategies where the Sharpe Ratio often suffices as a risk-adjusted metric, market making requires explicit consideration of inventory management, making PnL-MAP especially appropriate. MDD is also included to measure the strategy's resilience under adverse market conditions. Together, this combination ensures both generalizability and domain-specific relevance.

We referred to several rule-based methods used in previous studies (Spooner et al., 2018; Lim and Gorse, 2018), and also selected a

method from recent reinforcement learning research as a state-of-the-art (SOTA) comparison. The comparative Method are as follows:

- **Random**: This method randomly selects a quoting pair from all possible actions for placing an order at each quoting time (for example, in our problem setting, one of the 8 quoting pairs is randomly chosen).
- **Zero-tick**: Always place orders at best price(i.e. best ask & best bid), which corresponds to the action (0,0). This method implies that market makers will buy and sell at the most aggressive prices. Although this increases the chance of fulfillment, it may also pose a higher inventory risk.
- **Avellaneda-Stoikov(AS) model** (Avellaneda and Stoikov, 2008): A classic and effective strategy for managing inventory risk, by considering factors such as mid price($s$), inventory holding($q$), volatility($\sigma$), risk aversion parameter($\gamma$), remaining market trading time($T-t$), and order book liquidity($k$), finally derive the market reservation price($r$) and the optimal quote width($\delta_a+\delta_b$). The calculation formula are expressed in Eqs. (15) and (16):

$$r(s,q,t) = s - q\gamma\sigma^2(T-t) \tag{15}$$

$$\delta_a + \delta_b = \gamma\sigma^2(T-t) + \frac{2}{\gamma} ln(1+\frac{\gamma}{k}) \tag{16}$$

  The design of the reservation price ensures that the strategy will adjust the quoting skew according to the inventory position, reducing the chance of further increasing positions. Note that the parameters are fixed in this setting, where we set $\gamma = 0.1$, $\sigma = 3$, $k = 3$, and $q$ is the inventory ratio same as Section 3.3.
- **FinRL** (Liu et al., 2020): An open-source trading system that integrates several traditional RL algorithms tailored for financial applications. Given its wide adoption and representative nature, it is essential to conduct a comparative analysis between this established system and our proposed methodology.
- **MVMM** (He et al., 2023): MVMM (Multi-Agent Virtual Market Model) is a simulation framework that integrates multiple Generative Adversarial Networks (GANs) to generate realistic and diverse market conditions. By modeling interactions among simulated market participants, MVMM allows reinforcement learning agents to train in environments where their actions affect market states, overcoming limitations of historical-data-based training.
- **Alpha-AS** (Falces Marin et al., 2022): A DRL method which improves the AS model, where the RL agent determines the risk aversion parameter $\gamma$ and quoting skew in the AS model. The state comprises 32-dimensional data, including inventory and limit order book price and quantity information. There are 20 actions in total, which are the combinations of $\gamma \in \{0.01, 0.1, 0.2, 0.9\}$ and skew $\in \{-0.1, -0.05, 0, 0.05, 0.1\}$. The reward is the Asymmetric dampened PnL ($\eta = 0.3$), which is the realized PnL plus the loss part of the unrealized PnL, as shown in Eq. (17).

$$R(t) = \phi_t + q_t \Delta m_t - \eta \cdot max(0, q_t \Delta m_t) \tag{17}$$

### 4.3. Experiments on inventory penalty parameters

In this experiment, we aim to find the optimal hyperparameter $\eta$ for the inventory penalty term, in order to ensure our agent achieves the best balance between profitability and risk avoiding. We conduct this parameter experiment at the beginning to provide a stable setup for the subsequent ablation and method comparison experiments. By examining the experimental results, we observe the impact and correlation of different settings of the inventory penalty term hyperparameter $\eta$ on the training performance. We conducted experiments with ten different reward functions(refer to Section 3.3) with different $\eta$, and the values are range from 0.001 to 1. Fig. 5 displays our experimental results.

We used the environments from the training set to separately test the influence of $\eta$ in both call and put environments. The most basic setup was employed, that is, training and evaluation were conducted without incorporating the SSMCN embedding and CL mechanism. The left side of the figure shows the experimental results for the call option, while the right side depicts the results for the put option. The $X$-axis corresponds to different $\eta$ values, and the $Y$-axis corresponds to the magnitudes of different evaluation metrics. As the line chart of the experimental results indicates, the larger the $\eta$, the heavier the inventory penalty, leading the agent to lean towards reducing inventory during learning. Thus, we observe a decreasing trend from left to right for the MAP and INV metrics, which almost level off after surpassing 0.05.

Our goal is to identify the setting that best balances profit-risk performance. Therefore, we also present trends in profit amount (PnL) and the ratio of profit to inventory (PnL-MAP) in the graph. As seen in Fig. 5, the profit performance is not ideal with very small inventory penalty settings, regardless of whether it is in the call or put environment. In the early stages, as the inventory penalty increases, the profitability also rises. However, when the inventory penalty is too large, profitability starts to decline. Ultimately, we found that at $\eta = 0.05$, there is a lower average inventory holding, while maintaining better profitability. The PnL-MAP indicator also reveals higher values at $\eta = 0.05$. Therefore, this finding informs RQ1, suggesting that an inventory penalty of $\eta$ of 0.05 is optimal for balancing profit and risk to train an agent.

### 4.4. Multi-task ablation study for SSMCN

In this experiment, we aim to explore the effects of market context trained with various auxiliary tasks. We employ auxiliary tasks under different settings to train the SSMCN, subsequently inputting the market information, $X^{mkt}$, into the trained SSMCN to obtain diverse market contexts. These are then input into a downstream agent for training, yielding backtest results. We consider several settings: removing all market-related sentiment indicators (no mkt), removing all investor sentiment indicators of TXO (no TXO), removing all investor sentiment indicators of TXF (no TXF), removing the entire SSMCN module (no SSMCN), and using the full set of nine auxiliary tasks (All).

The purpose of dividing auxiliary tasks into three groups is to understand the contributions and impacts of different market investor sentiments. For instance, the first group, "no mkt", eliminates auxiliary tasks relating to market trend, volatility, and the put–call ratio, which we believe are crucial for market making. If a market context can depict daily market conditions such as strength or weakness, this may provide market making agents with the opportunity to learn whether the overall order placement on a given day should be more long or short, or to widen order placement in volatile markets to increase profit from bid–ask spread. The second and third groups seek to comprehend whether large market investor positions can indicate market sentiment, as mentioned in Section 3.2. The open positions of different roles in the market all have their significance and are one of the indicators investors refer to when trading.

Experimental results are displayed in Table 5, in which we average the daily test results over six months, and present the ablation study results for both call and put scenarios. Each column represents a different evaluation metric, while each row represents the auxiliary tasks used in different groups. In terms of the PnL-MAP indicator, the performance of using all auxiliary tasks was the best, both for calls and puts, implying that considering all auxiliary tasks collectively could help market-making agents achieve better profit performance.

In the results of the three groups of auxiliary tasks, when market information is removed, the PnL-MAP performance drops the most. We hypothesize this is due to the resultant market context lacking representation for market trends and volatility, making it harder for agents to adjust their strategies based on the market context. This could even make the strategies more unstable, leading to a larger MAP and INV in call scenarios, and poorer profits.

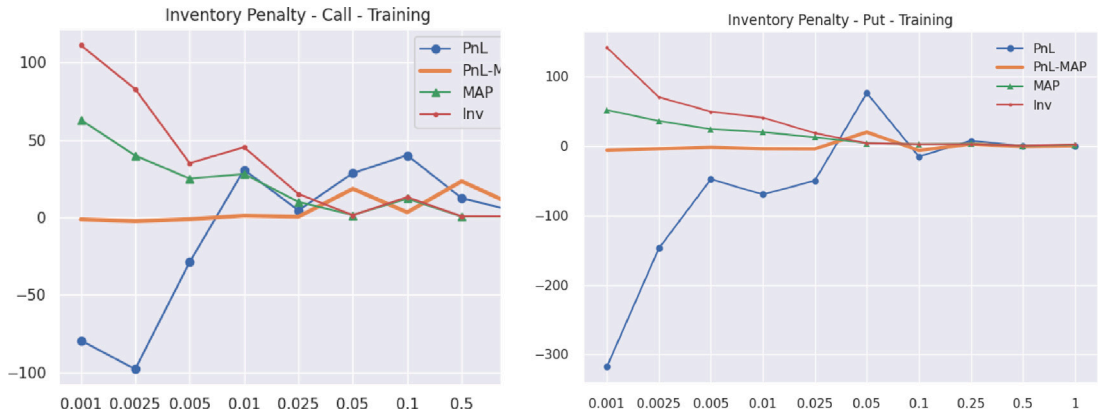


**Fig. 5.** Performance under different inventory penalty.

**Table 5**
Performance of different combinations of auxiliary tasks in the SSMCN.

| Period | Method | Evaluation metrics | | | | | |
|---|---|---|---|---|---|---|---|
| | | PnL | MAP | PnL-MAP | MDD | SR | INV |
| 201901 ~ 201906 | Call - All | 95.539 | **3.021** | **31.810** | **23.458** | **1.484** | **3.887** |
| | Call - no mkt | 33.991 | 39.567 | 0.914 | 187.845 | −0.153 | 65.788 |
| | Call - no TXO | 31.988 | 6.646 | 6.670 | 51.711 | 1.072 | 6.130 |
| | Call - no TXF | **101.892** | 6.974 | 15.109 | 34.237 | 1.069 | 11.719 |
| | Call - no SSMCN | 63.919 | 7.983 | 9.989 | 53.851 | 0.163 | 10.444 |
| | Put - All | 151.528 | 4.767 | **31.576** | 50.815 | 0.501 | 6.131 |
| | Put - no mkt | 128.856 | 9.079 | 16.556 | **33.404** | **1.873** | 15.697 |
| | Put - no TXO | 159.701 | 6.296 | 25.760 | 77.440 | 1.144 | 7.051 |
| | Put - no TXF | 62.460 | **2.348** | 26.884 | **33.404** | 0.598 | **3.125** |
| | Put - no SSMCN | **228.176** | 24.943 | 9.252 | 133.602 | 0.363 | 34.101 |

Next, the performance of removing TXO and TXF in PnL-MAP is similar, with the results being slightly better when removing TXF. In the call scenario, removing TXF led to higher PnL, while maintaining a reasonable level of inventory, hence performing better in PnL-MAP than removing TXO. However, in put scenarios, the results of removing TXF were relatively conservative, with both profits and inventory lower, resulting in similar PnL-MAP performance. However, regardless of whether TXF or TXO was removed, the PnL-MAP performance was worse compared to using all auxiliary tasks, indicating that market investor sentiment does enhance the performance of market context.

Finally, the performance of removing SSMCN shows that when the agent lacks reference market context information, the performance in put scenarios is extreme. Despite high profits, it is accompanied by excessively high MAP and INV, which is unacceptable in risk assessment, and the PnL-MAP is the worst among all configurations.

Overall, the experimental results reveal that the market context trained through all auxiliary tasks delivers stable results for both call and put scenarios, along with superior PnL-MAP performance, indicating its ability to increase stability and generalizability for downstream market making agents. This addresses RQ2, showing that the addition of auxiliary tasks in the SSMCN indeed contributes to performance improvement.

### *4.5. Comparison of continual learning methods*

The following experiment attempts to explore whether different CL methods can help the agent adapt to new environments. In the testing environment, for methods using CL, we perform environment selection and strategy fine-tuning for the agent every day and use the updated agent for backtesting. Note that we use the same pre-trained agent for updates every day, rather than updating the agent from the previous day. This is because we believe that the pre-trained agent already has a good basic market making ability, so it only needs to be fine-tuned according to the environment of the day. At the same time, it can also avoid the strategy being unable to adjust smoothly to the market of the day due to the significant difference in the environments of two days.

In this experiment, we used three types of CL methods as experimental methods, and the method without CL (i.e., the performance of the pre-trained agent) was used as the control group. The first setting, "no continual", uses the same pre-trained agent for all test days without further updates. The "similar context" setting applies our proposed continual learning update mechanism, where the agent selects Nenv historical environments with the most similar market contexts, based on Euclidean distance, for fine-tuning. The "latest" setting instead uses the most recent Nenv days for updates, without considering similarity. Finally, the "both" setting combines the two approaches by selecting both the most recent environments and the most similar ones.

We divided the 6-month testing environment into intervals of 2 months each, and presented the results in two tables. Table 6 shows the call option results, and Table 7 shows the put option results. According to the experimental results, whether it's call or put option data, using a similar market context as the selection mechanism for the update environment, the profit indicator PnL-MAP could be further improved in all intervals. In most intervals, it also maintains a smaller MAP. This suggests that fine-tuning strategies through our environment selection mechanism can effectively enhance the profitability of the market-making agent in new environments.

However, using recent environments for updates is evidently not a good practice. As can be seen, whether using the 'latest' or 'both' environment selection methods, the PnL-MAP performs worse than the model without CL. In terms of MDD, PnL, and SR, the 'latest' or 'both' environment also shows regression compared to the "similar" environment. This suggests that the market environment of a certain past period may be far from the current one. Updating arbitrarily without intelligence could make the model forget its original knowledge and learn in a biased direction. On the other hand, our method ensures that the market updates under similar circumstances. Compared to the original model, profits improved from January to February for both

**Table 6**
Performance of different continual learning methods on call options.

| Period | Method | Evaluation metrics | | | | | |
|---|---|---|---|---|---|---|---|
| | | PnL | MAP | PnL-MAP | MDD | SR | INV |
| 201901 ~ 201902 | no continual | 137.222 | 2.944 | 46.612 | **20.332** | 0.383 | 3.672 |
| | latest | 9.734 | 3.180 | 3.062 | 206.285 | −0.197 | 3.250 |
| | similar context | **150.919** | 2.498 | **60.411** | 106.959 | **1.211** | 2.094 |
| | both | 93.746 | **1.839** | 50.967 | 56.846 | 0.997 | **1.891** |
| 201903 ~ 201904 | no continual | **73.049** | 3.033 | 24.087 | **21.794** | **1.477** | 4.053 |
| | latest | 14.749 | 2.371 | 6.219 | 117.416 | −0.015 | 2.605 |
| | similar context | 65.697 | **1.129** | **58.170** | 29.266 | 1.075 | **1.500** |
| | both | 55.026 | 4.347 | 12.659 | 178.658 | 0.782 | 5.316 |
| 201905 ~ 201906 | no continual | **76.346** | 3.087 | 24.733 | **28.248** | **2.592** | 3.936 |
| | latest | −65.519 | 11.267 | −5.815 | 617.766 | 0.085 | 16.859 |
| | similar context | 47.856 | **1.814** | **26.383** | 89.038 | 0.953 | **1.538** |
| | both | −38.429 | 10.114 | −3.800 | 429.266 | 0.889 | 13.897 |

**Table 7**
Performance of different continual learning methods on put options.

| Period | Method | Evaluation metrics | | | | | |
|---|---|---|---|---|---|---|---|
| | | PnL | MAP | PnL-MAP | MDD | SR | INV |
| 201901 ~ 201902 | no continual | 199.055 | 4.942 | 40.279 | **44.485** | 0.216 | 6.703 |
| | latest | 178.516 | 4.511 | 39.575 | 148.551 | 0.731 | 6.266 |
| | similar context | **226.016** | 5.291 | **42.717** | 212.480 | **0.946** | 7.313 |
| | both | 98.223 | **3.039** | 32.316 | 112.324 | 0.738 | **3.484** |
| 201903 ~ 201904 | no continual | **173.701** | 4.717 | 36.823 | **55.763** | 0.324 | 4.934 |
| | latest | 100.923 | 4.024 | 25.078 | 136.287 | 0.470 | 5.263 |
| | similar context | 103.121 | **2.748** | **37.525** | 92.789 | **0.619** | **3.921** |
| | both | 69.982 | 3.566 | 19.625 | 129.532 | 0.527 | 4.447 |
| 201905 ~ 201906 | no continual | 81.830 | 4.642 | 17.627 | **52.197** | **0.964** | 6.756 |
| | latest | **136.689** | 4.659 | 29.340 | 237.000 | 0.673 | 5.974 |
| | similar context | 124.907 | **2.987** | **41.817** | 172.891 | 0.496 | **3.705** |
| | both | 66.260 | 3.146 | 21.062 | 156.240 | 0.572 | 3.910 |

call and put options, while in other months, a lower inventory risk is maintained. As a result, these findings answer RQ3. Using market context to select fine-tuning environments improves performance in test environments, as shown by higher PnL-MAP and lower inventory risks. In contrast, updating with only the most recent environments worsens results, highlighting that CL is effective only when guided by context-based selection.

### 4.6. System performance evaluation

In this experiment, we attempt to compare our method with other existing methods (see Tables 8 and 9). "Random" places quotes randomly without using market information. "Zero-tick" always quotes at the best bid and ask. "AS" is an analytical strategy based on adaptive quoting rules, while "Alpha-AS" integrates reinforcement learning into the AS framework. "MVMM" simulates market dynamics using multi-agent interactions and GAN-based scenario generation. "FinRL" is a general-purpose financial reinforcement learning framework trained on historical market features. The details of the methods are as described in Section 4.2.

Our method, even without the use of CL technology, already has a higher PnL-MAP than other control groups. In most of the intervals, it also gets better SR and MDD. The results indicate that our method's profit-making ability under unit risk is superior to other methods. However, it may not necessarily have the highest PnL performance. This is because we believe that controlling inventory risk is relatively important. In the call options, the remaining three control groups each achieved the highest profit in each interval, but these strategies have unstable profit performance. For example, Zero-tick has the highest profit from January to February but suffered losses from March to April. The other strategies often accompany higher MAPs when making profits. In put options, our method's profit performance is better, except for the interval from March to April losing to Zero-tick, it won over other control groups in other times. After adding CL, our method further improved its performance. For example, in all segments of the call options, it maintains the lowest MAP and the highest PnL-MAP. In the put options, it achieved the highest PnL and the lowest MAP in the intervals from January to February and from May to June, showing that our CL method can help the agent adapt to new environments and enhance stability.

When observing the profit-making and risk-taking abilities of the strategies, in addition to focusing on the three indicators of PnL, MAP, PnL-MAP, the SR, and MDD ratio are also common profit and risk indicators respectively. SR can represent the profit stability during market-making. In the call options, our method has a higher SR, while in the put options, it is slightly unstable. Except for January to February, it is higher than other control groups, the remaining months are lower than the AS model. However, overall, the profits of our method are quite stable. MDD is used to measure the maximum drawdown during market-making. In the call options, our method without CL has a lower MDD. Surprisingly, although the method with CL has a lower average inventory, it increases on MDD. We believe this is because this method is more active in participating in the market during the trading session, resulting in larger profit and loss changes.

Next, we analyze the performance of each control group. Among all the control groups, the best performance is the AS model. The AS model has good position control ability, can determine the offset of the order according to the inventory position, so it is relatively low in MAP, second only to our method. The profit part also has a good performance, obtaining the highest PnL in the period from May to June in the call options, and maintaining a certain level of performance on PnL-MAP. Alpha-AS is more aggressive. Although the PnL is higher than the AS model from January to April in the call options, the MAP is much higher. In the put options, the performance is even worse, almost at the bottom of PnL and MAP performance. We believe the possible reason is that sometimes the agent chooses more extreme parameters,

**Table 8**
Performance of comparison methods on call options.

| Period | Method | Evaluation metrics | | | | | |
|---|---|---|---|---|---|---|---|
| | | PnL | MAP | PnL-MAP | MDD | SR | INV |
| 201901 ~ 201902 | Random | 35.930 | 9.787 | 3.671 | 30.520 | 0.183 | 13.344 |
| | Zero-tick | **245.099** | 15.058 | 16.277 | 49.604 | 0.226 | 23.422 |
| | AS model | 183.826 | 7.007 | 26.236 | 20.893 | 0.277 | 11.188 |
| | MVMM | −9.088 | 7.838 | −1.159 | 414.184 | 0.349 | 7.281 |
| | FinRL | 157.330 | 11.720 | 13.424 | 429.570 | 0.250 | 9.910 |
| | Alpha-AS | 188.892 | 40.427 | 4.672 | 175.107 | 0.284 | 24.286 |
| | MM-CRL (no CL) | 137.222 | 2.944 | 46.612 | **20.332** | 0.383 | 3.672 |
| | MM-CRL | 150.919 | **2.498** | **60.411** | 106.959 | **1.211** | **2.094** |
| 201903 ~ 201904 | Random | 12.414 | 8.396 | 1.479 | 45.009 | 0.861 | 13.158 |
| | Zero-tick | −80.382 | 18.907 | −4.252 | 72.452 | 0.866 | 31.618 |
| | AS model | 32.704 | 6.500 | 5.031 | 32.086 | 0.158 | 12.908 |
| | MVMM | 34.266 | 5.738 | 5.971 | 213.681 | 0.363 | 4.711 |
| | FinRL | −7.210 | 8.440 | −0.854 | 350.830 | 0.130 | 7.800 |
| | Alpha-AS | **83.141** | 41.010 | 2.027 | 161.051 | 1.085 | 33.961 |
| | MM-CRL (no CL) | 73.049 | 3.033 | 24.087 | **21.794** | **1.477** | 4.053 |
| | MM-CRL | 65.697 | **1.129** | **58.170** | 29.266 | 1.075 | **1.500** |
| 201905 ~ 201906 | Random | −62.016 | 9.726 | −6.376 | 60.830 | 1.486 | 15.641 |
| | Zero-tick | −43.782 | 18.564 | −2.358 | 106.831 | 1.285 | 29.359 |
| | AS model | **81.404** | 8.002 | 10.173 | 43.385 | 1.990 | 14.897 |
| | MVMM | −16.779 | 9.318 | −1.801 | 432.016 | 0.181 | 9.974 |
| | FinRL | −15.530 | 11.010 | −1.411 | 482.130 | 0.070 | 9.690 |
| | Alpha-AS | 30.320 | 38.047 | 0.797 | 198.046 | 0.019 | 33.679 |
| | MM-CRL (no CL) | 76.346 | 3.087 | 24.733 | **28.248** | **2.592** | 3.936 |
| | MM-CRL | 47.856 | **1.814** | **26.383** | 89.038 | 0.953 | **1.538** |

**Table 9**
Performance of comparison methods on put options.

| Period | Method | Evaluation metrics | | | | | |
|---|---|---|---|---|---|---|---|
| | | PnL | MAP | PnL-MAP | MDD | SR | INV |
| 201901 ~ 201902 | Random | 93.734 | 9.972 | 9.400 | 38.313 | 0.113 | 14.031 |
| | Zero-tick | 197.836 | 15.096 | 13.106 | 60.041 | 0.179 | 21.344 |
| | AS model | 130.742 | 7.723 | 16.929 | **34.001** | 0.232 | 11.828 |
| | Alpha-AS | 25.559 | 23.349 | 1.095 | 113.529 | 0.072 | 13.571 |
| | MM-CRL (no CL) | 199.055 | 4.942 | 40.279 | 44.485 | 0.216 | **6.703** |
| | MM-CRL | **226.016** | **5.291** | **42.717** | 212.480 | **0.946** | 7.313 |
| 201903 ~ 201904 | Random | 108.266 | 9.008 | 12.019 | 62.666 | 0.586 | 13.684 |
| | Zero-tick | **237.404** | 15.163 | 15.657 | 98.622 | 0.554 | 22.184 |
| | AS model | 179.746 | 6.354 | 28.289 | **52.445** | **0.979** | 11.250 |
| | Alpha-AS | 38.749 | 19.963 | 1.941 | 98.390 | 0.164 | 17.618 |
| | MM-CRL (no CL) | 173.701 | 4.717 | 36.823 | 55.763 | 0.324 | 4.934 |
| | MM-CRL | 103.121 | **2.748** | **37.525** | 92.789 | 0.619 | **3.921** |
| 201905 ~ 201906 | Random | −7.785 | 10.391 | −0.749 | 84.193 | 1.218 | 14.295 |
| | Zero-tick | 40.939 | 17.349 | 2.360 | 120.266 | 1.278 | 26.654 |
| | AS model | 80.824 | 8.983 | 8.998 | 64.071 | **2.781** | 15.218 |
| | Alpha-AS | 55.007 | 19.407 | 2.834 | 114.248 | 0.077 | 20.987 |
| | MM-CRL (no CL) | 81.830 | 4.642 | 17.627 | **52.197** | 0.964 | 6.756 |
| | MM-CRL | **124.907** | **2.987** | **41.817** | 172.891 | 0.496 | **3.705** |

causing the calculated price to be far different from the market price, and therefore cannot smoothly reduce the position. MVMM and FinRL exhibit less stable profitability and larger inventory risks compared to our method, suggesting that these methods may be less well-suited for the specific demands of market making. Also, Zero-tick is the most aggressive method among the methods. It can be seen that the MAP is mostly higher, and the PnL performance is very unstable. Although it has the highest PnL from May to June in the call options and from March to April in the put options, it is almost at the bottom at other times. To evaluate the statistical significance of our method's improvement, we conducted a paired t-test comparing the PnL-MAP of MM-CRL and the AS model, the strongest-performing baseline, and resulted in a $p$-value of 0.0003, confirming that the difference is statistically significant. Specifically, MM-CRL achieved a mean PnL-MAP of 30.50 (95% CI: [23.65, 37.36]), while the AS model achieved 13.81 (95% CI: [7.82, 19.81]), further demonstrating the robustness and superiority of our approach.

Finally, it is worth mentioning that our method performs excellently in INV, indicating that it can reduce the position to almost zero before the close, which can reduce hedging costs and overnight risks. The overall results also show that our method is competitive and provide clear evidence to answer RQ4, our proposed method outperforms other control groups. Not only can it maintain considerable profit-making ability in various commodities and time segments, but it also has excellent inventory control ability.

### 4.7. Profitability evaluation for additional dataset

To further assess the generalizability of our approach, we conducted additional experiments on a new dataset. We utilized tick data from Taiwan Electronics Options (TEO), listed on the TAIFEX. TEO is the second most actively traded product in the Taiwan options market. The training and testing periods were selected to match those used for the TXO. The experimental results are presented in Table 10, which displays the testing outcomes for the TEO and the average daily test results over six-month periods.

Similar to the main results, MM-CRL achieves the highest PnL-MAP and SR, demonstrating risk-adjusted returns compared to all baselines.

**Table 10**
Performance of comparison methods on TEO options.

| Period | Method | Evaluation metrics | | | | | |
|---|---|---|---|---|---|---|---|
| | | PnL | MAP | PnL-MAP | MDD | SR | INV |
| 201901 ~ 201906 | Random | 8.258 | 9.856 | 0.838 | 374.706 | 0.252 | 15.995 |
| | Zero-tick | 28.268 | 17.654 | 1.601 | 655.425 | 0.184 | 28.404 |
| | AS model | **94.495** | 7.186 | 13.149 | 296.237 | 0.324 | 13.114 |
| | MVMM | 45.088 | 20.468 | 10.356 | 736.825 | 0.237 | 33.889 |
| | FinRL | 45.703 | **2.793** | 16.256 | **96.314** | 0.144 | **2.801** |
| | MM-CRL | 72.160 | 3.471 | **20.721** | 118.506 | **0.326** | 5.767 |

Notably, AS model achieves the highest raw profitability, reflecting a more aggressive trading style, while FinRL shows the lowest MAP and MDD, reflecting a more conservative and risk-averse approach. The performance of MMCRL indicates that it strikes a well-balanced trade-off between profitability, risk control, and inventory management, effectively adapting to the unique market dynamics of TEO options.

In conclusion, the experimental results highlight the ability of our method to balance profitability and risk control in the TEO market. This further substantiates the robustness and adaptability of our framework across different asset classes and market conditions. The consistent outperformance on key risk-adjusted metrics across both the TXO and TEO datasets underscores the broad applicability and practical relevance of our approach in addressing real-world market making challenges. These results thus provide a direct answer to RQ5, confirming the generalizability of our method to an additional dataset.

After addressing all the proposed research questions (RQ1–RQ5), we are able to refine the overall objective of this study. Specifically, the primary goal of this research is twofold: (i) to tackle the problem of concept drift in dynamic market environments, and (ii) to enhance the generalization ability of the proposed method across varying conditions. By introducing a self-supervised market context network with multiple auxiliary tasks, we are able to extract informative features from heterogeneous market sentiment indicators, thereby enabling the agent to adjust its decisions more effectively in response to different market conditions. Furthermore, continual learning is incorporated to mitigate concept drift by selecting and leveraging appropriate historical environments. These mechanisms allow the proposed framework to achieve more robust adaptation and ultimately support more effective financial decision-making.

## 5. Conclusion

In this study, we propose a novel Contextual Reinforcement Learning framework for market making. This framework uses financial market sentiment indicators as context information and employs multi-environment training to enable the reinforcement learning agent to adjust market making quoting decisions according to market conditions, enhancing the model's generalization ability. Additionally, we implement a self-supervised representation learning method. This method uses multiple auxiliary tasks and incorporates financial domain knowledge into feature learning, which greatly helps downstream market making tasks and continual learning mechanism. To address the concept drift issue, we introduce a continual learning method to continuously adjust the policy. This method uses the similarity of market contexts to sample environments from historical data. By retraining the agent with environments with similar market conditions, the model's performance in new environments can be ensured to be more stable. Finally, we verify the performance of our methods through experiments using real option trading data. These experiments have demonstrated the power of our methods. Compared with the best control group, our agent's PnL-MAP has increased 34.5 on call options and 22.6 on put options, indicates that our method possesses higher profitability potential and can achieve lower inventory risk.

In future research, exploring more interpretable policy designs by analyzing how the agent adapts its quoting behavior under different market contexts could help improve the practical usability of the framework. For the problem of options market making, we can try to incorporate hedging mechanisms, which consider the profit and loss of both options and futures positions to reflect more realistic market making behavior. We can also attempt to incorporate a hierarchical RL structure, where a manager's role determines how to switch market making strategies based on market conditions. The modeling of high-frequency sentiment and market microstructure can also be more refined, as well as designing more realistic trade determination mechanisms and order placement behaviors. In addition, future work could also benefit from interdisciplinary insights, such as applying blockchain mechanisms to enhance transparency and auditability of RL agents (Zhu et al., 2023), integrating hybrid modeling approaches for improved robustness under uncertainty (Zhu et al., 2024), and leveraging label noise filtering techniques to ensure data quality and interpretability (Xu et al., 2025).

## CRediT authorship contribution statement

**Pin-Yao Wen:** Writing – original draft, Validation, Methodology, Formal analysis, Data curation. **Szu-Hao Huang:** Supervision, Resources, Funding acquisition, Conceptualization. **Chiao-Ting Chen:** Writing – review & editing, Project administration. **Yi-Tang Fang:** Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

The data that has been used is confidential.

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